O-I-507D
February 5, 1974
SUPERSEDING
Fed. Spec. O-I-507C
April 30, 1970

FEDERAL SPECIFICATION

INSECTICIDE, ALLETHRIN OR PYRETHRIN, (AEROSOL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 This specification covers one aerosol insecticide formulation containing synergized allethrin or pyrethrins in a 12-ounce dispenser complete with valve.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

O-P-921 - Pyrethrum Extract.
BB-F-1421 - Fluorocarbon Refrigerants.
TT-W-572 - Wood Preservative: Water-Repellent.
VV-K-220 - Kerosene, Deodorized.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

FSC 6840

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-2427 - **Box,** Ammunition Packing; Wood, Nailed.
MIL-P-15011 - **Pallet,** Material Handling, Wood, Post Construction, 4-Way Entry.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Code of Federal Regulations.

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171-179.")

Federal Environmental Pesticide Control Act.

(The Federal Environmental Pesticide Control Act of 1972 is under the jurisdiction of the Office of Pesticides Programs, Environmental Protection Agency, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee.

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standard:

D1193-70 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Association of Official Analytical Chemists (AOAC).

Official Methods of Analysis, 10th Edition, 1965.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Incorporated, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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Soap and Sanitary Chemicals, Vol. XXIX, No. 2, February, 1953.

(Soap and Sanitary Chemicals is published by the MacNair-Dorland Company, 101 West 31st Street, New York, New York 10001.)

U. S. Department of Agriculture Publication:

ARS-33-5 - A Method of Determining Particle Size of Liquified Gas Aerosols (March, 1955).

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Aerosol insecticide formulation composition. The aerosol insecticide formulation shall conform to the composition of table I. The individual ingredients shall conform to the applicable reference paragraph specified therein.

3.1.1 Allethrin, technical. Allethrin, technical shall conform to the composition of table II when tested as specified in 4.2.4.1.

3.1.2 Pyrethrum extract. Pyrethrum extract shall conform to O-P-921 except that the pyrethrins content shall be no less than 20 percent by weight when tested as specified in O-P-921.

Table I. Composition of aerosol insecticide formulation

Ingredient	Percent by weight	Reference paragraph
Insecticide, minimum:		
Allethrin, technical, or	2.8	3.1.1
Pyrethrum extract	3.0	3.1.2
Synergist, minimum:		3.1.3
Piperonyl butoxide, technical, or	1.4	
"Tropital," technical	1.4	
Dichlorodifluoromethane	43.5 ± 1.0	3.1.4
Trichloromonofluoromethane	43.5 ± 1.0	3.1.5
Deodorized kerosene	Remainder	3.1.6

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Table II. Allethrin, technical composition

Ingredient	Percent by weight
Allethrin (actual), minimum	90.0
Chrysanthemumic acid plus chrysanthemumic anhydride, maximum	4.0
Chrysanthemumic acid chloride, maximum	0.2
Dichlorodifluoromethane insolubles, maximum	0.05

3.1.3 Synergist.

3.1.3.1 Physical characteristics. The synergist shall conform to the applicable physical characteristics of table III when tested as specified therein.

Table III. Physical characteristics of synergists

Synergist	Specific gravity*	Refractive index**
Piperonyl butoxide, technical	1.04 to 1.07 (20°C)	1.480 to 1.500 (20°C)
"Tropital," technical	1.06 to 1.09 (20°C)	1.474 to 1.490 (25°C)

*Determined by any suitable method accurate to 0.001.
**Determined by Abbe refractometer or other suitable instrument.

3.1.3.2 Chemical characteristics.

(a) Piperonyl butoxide, technical. The piperonyl butoxide, technical shall contain no less than 80 percent by weight of alpha[2-(2-butoxyethoxy)ethoxy]-4,5-(methylenedioxy)-2-propyltoluene when tested as specified in 4.2.4.2.

(b) "Tropital," technical. The "Tropital," technical shall contain no less than 90 percent by weight of piperonal bis[2-(2-butoxyethoxy)ethyl] acetal when tested as specified in 4.2.4.3.

3.1.4 Dichlorodifluoromethane. The dichlorodifluoromethane shall conform to the requirements of type 12 of BB-F-1421 when tested as specified therein.

3.1.5 Trichloromonofluoromethane. The trichloromonofluoromethane shall conform to the requirements of type 11 of BB-F-1421 when tested as specified therein.

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3.1.6 Deodorized kerosene. The deodorized kerosene shall conform to the requirements of VV-K-220 when tested as specified therein.

3.2 Allethrin in aerosol insecticide formulation. When allethrin is used as the insecticide, the aerosol insecticide formulation shall contain no less than 2.5 percent by weight of actual allethrin based on the analysis of the allethrin when tested as specified in 4.2.4.5.

3.3 Pyrethins in aerosol insecticide formulation. When pyrethrum extract is used as the insecticide, the aerosol insecticide formulation shall contain no less than 0.6 percent by weight of pyrethrins when tested as specified in 4.2.4.6.

3.4 Water content in aerosol insecticide formulation. After the dispenser has been filled with the aerosol insecticide formulation, the formulation shall contain no more than 80 parts per million of water when tested as specified in 4.2.4.7. In order to minimize water content, it is recommended that filling be carried out at a relative humidity equivalent to no more than 30 percent at 70° F.

3.5 Net weight of contents. Each dispenser shall be filled with sufficient aerosol insecticide formulation so that the dispenser shall contain no less than 12 ounces of formulation when tested as specified in 4.2.4.4.

3.6 Dispenser and valve. The dispenser container shall be made from number 25 electrolytic tin plate and shall conform to type IX, class 2, protective coating plan B with side seam striped of PPP-C-96. The container shall have a capacity of no less than 25 cubic inches. The container shall be supplied with a dispensing valve having the valve performance characteristics specified in table IV. The valve shall be fitted with a finger operated side discharge plastic button. The direction of discharge shall be indicated either by the shape of the button or by a mark on it. The valve and button shall be constructed of materials which are not affected by the formulation.

Table IV. Valve performance characteristics

Characteristic	Requirement	Test paragraph
Delivery rate at 80° F	0.75 to 1.35 g per second	4.2.4.8
Particle mass median diameter:		4.2.4.9
80 percent	30 microns or less	
0 percent	50 microns or above	
Leakage in 48 hours, maximum	0.5 ml	4.2.4.10

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3.7 Protective cap. The dispensing valve shall be protected against actuation by a snug-fitting, double lacquer coated, protective cap made of 85-pound can makers quality MR-T_2 or MR-T_3 black plate. The cap shall provide a clearance of no less than 1/8 inch directly above the valve and shall completely cover the can end containing the valve assembly. The thickness of the cap directly over the valve shall be no less than 9 mils. The exterior protective coating shall be as specified for the dispenser container.

3.8 Workmanship. The aerosol insecticide formulation shall be mixed by a suitable process to produce a homogeneous mixture. The dispenser shall be clean and uniform and free of any defects which might impair its utility such as sediment, metal particles, or other foreign material in the container, eduction tube, or valve assembly. The exterior orifice shall be symmetrical and free of ragged edges. The exterior orifice, if drilled, shall be symmetrical in direct alignment with the angle of discharge.

3.9 Labeling. Labeling data shall be legibly lithographed on the dispenser container with a yellow, weather-resistant noncorrosive ink. Each dispenser shall be labeled in compliance with the Federal Environmental Pesticide Control Act. The label shall be accepted by and registered with the Environmental Protection Agency. The label shall include the following information:

(a) The name and percent by weight of each active ingredient.
(b) The name and percent by weight of each inactive ingredient.
(c) Directions for use:

DIRECTIONS FOR USE

For use against mosquitoes, flies, and other small flying insects. If possible, close space to be treated. Release 3 feet from any surface, holding the dispenser above the head. Moving the dispenser about while spraying aids in dispersion. Dispense only for time indicated. USE SPARINGLY. Seven seconds of spraying is ample for each 1,000 cubic feet (10 feet by 10 feet by 10 feet). It is not necessary to spray directly on insects but merely spray into the air since the aerosol will disperse throughout the enclosed space. Temporarily effective in sheltered outdoor areas.

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(d) Precautionary information:

WARNING: Contents under pressure. Do not puncture. Do not use or store near heat or open flame. Exposure to temperatures above 130° F may cause bursting. Never throw container into fire or incinerator.

CAUTION

Keep out of reach of children.
Harmful if swallowed.
Avoid spraying directly into eyes.
Avoid inhalation of aerosal mist.
Do not remain in treated areas. Ventilate the area after treatment is completed.
Never spray directly on plastics.
Avoid contamination of feed and food stuffs.
Remove pets, birds, and cover fish aquariums before spraying.
If used in food areas, food should be removed or covered during treatment. All food processing surfaces should be covered during treatment or thoroughly cleaned before using. The insecticide should be used in food processing areas only when the facility is not in operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dispensers that have been filled with aerosol insecticide formulation which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of filled dispenser. Sampling for examination of filled dispensers shall be conducted in accordance with MIL-STD-105, level I, using an AQL of 2.5 percent defective.

4.2.2.2 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105, using an AQL of 4.0 percent defective.

4.2.2.3 For test. Sampling for test shall be conducted in accordance with MIL-STD-105, level S-3, using an AQL of 2.5 percent defective.

4.2.3 Inspection procedure.

4.2.3.1 For composition of aerosol insecticide formulation. Weight records of the amount and lot number of each ingredient used to produce each lot of aerosol insecticide formulation shall be kept for the information of an inspector.

4.2.3.2 For examination of filled dispensers. The sample unit shall be one filled dispenser. Sample dispensers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Dispenser corroded, bent or distorted.
(b) Dispenser container not as specified.
(c) Valve discharge button not as specified.
(d) Protective cap not as specified.
(e) Labeling incorrect, missing, or illegible.

4.2.3.3 For examination of preparation for delivery. The sample unit shall be one filled shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container or arrangement thereof not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Fiberboard packing aids not as specified.
(f) Strapping or taping not as specified.
(g) Wood preservation not as specified (level A only).
(h) Marking incorrect, missing, or illegible.
(i) Palletization not as specified (when required).

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4.2.3.4 For test. Each sample dispenser taken in 4.2.2.3 shall be tested for delivery rate at 80° F, particle mass median diameter, leakage, and net weight of contents using an AQL of 2.5 percent defective. Three of the sample dispensers shall also be tested for water content. In addition, portions of the nonvolatile matter from these three sample dispensers shall be separately tested for percent allethrin or pyrethrins, as applicable. Failure of any of the three sample dispensers to pass the water content test or the allethrin or pyrethrins test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Allethrin analysis. Determine the percent by weight of actual allethrin and the percent by weight of chrysanthemumic derivatives in the allethrin, technical in accordance with AOAC methods 4.181 through 4.186. Determine the percent by weight of dichlorodifluoromethane insolubles in the allethrin, technical in accordance with the applicable test procedure specified in O-P-921.

4.2.4.2 Piperonyl butoxide, technical. Determine the percent by weight of alpha[2-(2-butoxyethoxy)ethoxy]-4,5-(methylenedioxy)-2-propyltoluene in piperonyl butoxide, technical in accordance with AOAC methods 4.256 through 4.259.

4.2.4.3 "Tropital," technical.

(a) "Tropital" solution. Prepare a solution of the technical "Tropital" specimen in a 50 percent benzene-50 percent hexane mixture so that the concentration of "Tropital" is 10 micrograms per milliliter (10 mg per liter).

(b) Procedure, part 1. Using a gas chromatograph equipped with an electron capture detector and chromatographic conditions conforming to table V, prepare a dosage-response curve of analyzed reagent grade piperonal in 50 percent benzene-50 percent hexane solutions having concentrations of 1.25, 2.50, 5.00, and 10.00 micrograms per milliliter. (The resolution time for the piperonal peak after time of injection is approximately 3.25 minutes under these conditions.) If a gas chromatograph of low sensitivity is used, higher concentrations of the specimen and piperonal may be used. Inject 1 to 10 microliters of

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Table V. Chromatographic conditions

Characteristic	Condition
Column material	Aluminum
Column dimensions	24 in. long, 1/8 in. diameter
Column coating (see 6.5)	10% "Carbowax 20M TPA"
Column support (see 6.5)	60/80 mesh "Chromasorb A/W DMCS"
Column temperature, °C	174
Gas	Nitrogen
Gas flow rate, ml per minute	20
Injection temperature, °C	210 to 225

the "Tropital" solution prepared as specified in (a) into the chromatograph and calculate the micrograms of unreacted piperonal present by using the dosage-response curve for piperonal previously prepared.

(c) Procedure, part 2. To convert the acetal in "Tropital" to piperonal and butylcarbitol, transfer 50 ml of the "Tropital" solution prepared as specified in (a) to a separatory funnel. Add 10 ml of 1.0N hydrochloric acid, stopper, and shake for 2 minutes (about 350 times). Let the phases separate and drain away the lower hydrochloric acid layer. Wash the benzene-hexane layer once by adding 10 to 15 ml of water and shaking 25 times. Drain off the wash water. Dry the benzene-hexane layer over anhydrous sodium sulfate. Inject 1 to 10 microliters into the chromatograph and determine the total micrograms of piperonal by comparison with the piperonal dosage-response curve.

(d) Calculation. Calculate the percent "Tropital," as piperonal bis[2-(2-butoxyethoxy)ethyl] acetal, as follows:

$$\text{Percent "Tropital"} = \frac{304(A - B)}{C}$$

where: A = Micrograms of piperonal found in (c),
B = Micrograms of piperonal found in (b), and
C = Micrograms of "Tropital" specimen in the "Tropital" solution injected in (c).

4.2.4.4 Net weight of contents and percent nonvatile matter.

(a) Percent nonvolatile matter. Weigh a dried 125-ml suction flask fitted with a 6- to 7-inch capillary tube having a 0.005- to 0.01-millimeter (mm) bore which has been inserted in a No. 5 rubber stopper.

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Remove the capillary tube from the flask and place the flask in dry ice or a low temperature cabinet for 10 minutes. Using a can-puncturing device as described in the water content determination (see 4.2.4.7) with a 1/4-inch flare nut with 6 inches of 1/4-inch copper tubing inserted in a No. 5 rubber stopper, attach the thief assembly to the filled dispenser can, puncture, and weigh the complete unit. Slowly add 30 to 50 g of the aerosol insecticide formulation from the can to the suction flask while the flask is in the dry ice or the low temperature cabinet. Reweigh the can and assembly and calculate the amount of formulation added to the flask by difference. Remove the flask from the cold source and replace the stopper and capillary tube. Allow the flask and contents to come to room temperature. Connect the flask to a vacuum pump capable of producing a vacuum of 0.01 mm of mercury and slowly increase the vacuum until there is no danger of losing part of the contents by boiling. Increase the vacuum to the full capacity of the pump and then immerse the flask in a water bath maintained at 38° C for five minutes. Remove the flask, dry the outside, and weigh. Connect the flask to the vacuum and immerse in the water bath for 1 minute. Dry the flask and reweigh. The first and second weighings should agree within 0.1 g. If a greater difference is obtained, repeat the procedure by placing the flask in the water bath for another minute. Retain the nonvolatile residue for the allethrin or pyrethrins determination, as applicable. Calculate the percent nonvolatile matter as follows:

$$\text{Percent nonvolatile matter} = \frac{100(A - B)}{W}$$

where: A = Weight of nonvolatile matter and flask in grams,
B = Weight of flask in grams, and
W = Weight of formulation added to flask in grams.

NOTE: The value obtained will be approximately 0.2 percent below the actual value.

(b) Net weight of contents. Discharge the remaining contents in the can and place the can in an oven at 80° C for 1 hour with the valve open. The loss in weight from the original weight of the filled can is the net weight of contents.

4.2.4.5 Allethrin in aerosol insecticide formulation. Determine the percent actual allethrin in the nonvolatile residue from 4.2.4.4(a) in accordance with AOAC methods 4.181 through 4.186. Calculate the percent allethrin in the aerosol insecticide formulation as follows:

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$$\text{Percent allethrin} = \frac{AB}{100}$$

where: A = Percent nonvolatile matter calculated in 4.2.4.4(a), and
B = Percent allethrin in the nonvolatile matter.

4.2.4.6 Pyrethrins in aerosol insecticide formulation. Determine the percent pyrethrins in the nonvolatile residue from 4.2.4.4(a) in accordance with the applicable test procedures specified in O-P-921. Calculate the percent pyrethrins in the aerosol insecticide formulation as follows:

$$\text{Percent pyrethrins} = \frac{AB}{100}$$

where: A = Percent nonvolatile matter calculated in 4.2.4.4(a), and
B = Percent pyrethrins in the nonvolatile matter.

4.2.4.7 Water content in aerosol insecticide formulation. Determine water content in accordance with the method described by R.A. Fulton and R.P. Gelardo in "A Method for Determining Water in Insecticidal Aerosols," Soap and Sanitary Chemicals, Volume XXIX, No. 2, pp. 145, 147, and 159, February, 1953.

4.2.4.8 Delivery rate of 80° F. Hold the dispenser upright and spray for 2 seconds to fill the eduction tube. Weigh the dispenser to the nearest 0.1 g. Submerge the dispenser in water bath maintained at 80.0° ± 0.5° F at least 1 inch below the water surface for 15 minutes. Remove the dispenser from the water bath with tongs and dry the dispenser with a towel. Spray the dispenser in one continuous burst for 10 seconds. Reweigh the dispenser to the nearest 0.1 g. Calculate the delivery rate in grams per second by dividing the loss in weight by 10. Repeat the procedure two more times and report the average of three results. The difference between the minimum and maximum rates shall not exceed 0.2 g per second.

4.2.4.9 Particle mass median diameter. Determine the particle mass median diameter in accordance with the method described by A.H. Yeomans, entitled "A Method of Determining Particle Size of Liquid Gas Aerosols," U.S. Department of Agriculture, ARS-33-5, March, 1955.

4.2.4.10 Leakage. Dispense ten separate 1-second applications of the aerosol formulation from each dispenser container selected for leakage testing. Then immerse the dispenser container in a deep beaker of water and invert a 1-ounce, wide-mouth vial, calibrated to read in milliliters, full of water over the valve of the

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container and allow to stand for 48 hours at 25° ± 2° C. Record the gas volume in the calibrated vial as the amount of leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Twenty-four filled dispensers shall be packed in six rows of four dispensers per row in a snug-fitting box conforming to class 2, style 4-1/2 of PPP-B-621; type overseas, style I or J of PPP-B-601; or type III, style optional of PPP-B-585. The box shall be designed for a type 2 load. Each dispenser shall be placed in a snug-fitting cell formed by use of half-slotted fiberboard partitions of the same height as the dispenser ± 1/8 inch. All inside faces of the box shall be lined with fiberboard. Additional fiberboard sheets shall be added where needed to assure a tight pack. The fiberboard partitions and liners shall be formed from material conforming to grade W5c or W6c of PPP-F-320. Unless there is provision for preservation in the applicable box specification, the wood box or the finished, unpainted wood parts thereof shall be preserved by total immersion for no less than 3 minutes in water-repellent wood preservative conforming to composition A of TT-W-572. Prior to packing the dispensers in the box, the box shall be dry when tested for dryness of preservative as specified in MIL-B-2427. The filled box shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification.

5.1.2 Level B.

5.1.2.1 For military activities. Twenty-four filled dispensers shall be packed in six rows of four dispensers per row in a snug-fitting box conforming to style RSC, grade W5c of PPP-B-636. Each dispenser shall be placed in a snug-fitting cell formed by use of half-slotted fiberboard partitions of the same height as the dispensers ± 1/8 inch. Fiberboard pads shall be placed against the top and bottom faces of each box. Additional pads shall be added where needed to assure a tight pack. Fiberboard partitions and pads shall be formed from material conforming to the minimum requirements of type CF, class domestic, variety and grade optional of PPP-F-320. The filled box shall be closed and reinforced as specified in the appendix to PPP-B-636.

5.1.2.2 For civil agencies. Twenty-four filled dispensers shall be packed in six rows of four dispensers per row in a close-fitting box

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conforming to class domestic of PPP-B-636. Slotted partitions shall be used to separate each dispenser. The partitions shall be made of the same material as used for the fiberboard box. The box shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C. Twenty-four filled dispensers shall be packed in a fiberboard box in compliance with Department of Transportation (DOT) regulations and Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable to the mode of transportation. Boxes shall be capable of being stacked and supporting superimposed loads during shipment and shall assure safe delivery to destination without damage to contents.

5.2 Marking. Marking shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military activities. In addition, shipping containers shall be marked in accordance with DOT regulations.

5.3 Palletization. When specified in the contract or order (see 6.2), uniform quantities of shipping containers shall be palletized for level A and level B shipments in accordance with the requirements for load type I of MIL-STD-147 using a pallet conforming to style 1, type I, class 1 of MIL-P-15011. For level A shipments, galvanized steel strapping shall be used and wood storage aids, including wood caps, shall be treated as specified for the pallet in MIL-P-15011. For level B shipments, a fiberboard cap shall be used and wood parts used in palletization need not be treated as specified for the pallets used for level A shipments. For level C shipments, any palletization shall be permitted provided that it conforms to Uniform Freight Classification rules or National Motor Freight Classification rules and that it permits stacking of palletized shipping containers in tiers 10 feet high without damage.

6. NOTES

6.1 Intended use. The aerosol insecticide covered by this specification is intended for indoor use against mosquitoes, flies, and other small flying insects. It may also be used for control of mosquitoes and flies in nonpressurized aircraft.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Palletization, if required (see 5.3).

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Chromatographic coating and support. "Carbowax 20M TPA" may be obtained from Union Carbide Corporation, Chemicals and Plastics Division, 270 Park Avenue, New York, New York 10017. "Chromasorb A/W DMSC" may be obtained from Johns-Manville, 22 East 40th Street, New York, New York 10016.

MILITARY INTEREST:

Custodians

Army - EA
Navy - SH
Air Force - 68

Review activities:

Army - MD, GL
Navy - MS, YD
DSA - GS

User activities:

Army - SM
Navy - MC

Preparing activity:

Army - EA

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR - ARS
EPA - PRD
GSA - FSS
HEW - FDA
VA - DMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

O-I-507D **Insecticide, Allethrin or Pyrethrin, (Aerosol)**

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Commander
Edgewood Arsenal
Attn: SAREA-TS-S
Aberdeen Proving Ground, MD 21010

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